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Final Examination, Business 33801:

(Monday, 23 January 2023)

Read all questions carefully and write answers in the space provided. If you run out of space, you may write on the back of the exam pages; if you run out of space on the back of the exam, you may add additional pages of paper. Graphs must be thoroughly labeled for full credit. Show your work and explain all equations on numeric questions. Be concise – irrelevant remarks will not count for anything. Be careful to budget your time between various parts, taking into account the points allocated to each question (the number in parentheses); these points vary depending upon the question.

There are 13 questions with a total of 180 points possible. You have 180 minutes for the examination, so assign about one minute per point. Allocate your time wisely.

YOU MUST READ AND SIGN THE HONOR STATEMENT BELOW IN ORDER TO RECEIVE CREDIT FOR THE EXAMINATION.

I pledge my honor that I have not violated the Honor Code during this examination.

Sign your name here.

True-False-Uncertain (30 points total)

Indicate whether the italicized portion of each statement is True, False, or Uncertain and EXPLAIN your answer. Read each question carefully. You may take the non-italicized portion of the statement as true. **No credit will be given for answers without explanation.**

Question 1 (6 points) True, False, Uncertain? [Explain your answer.] The market for interchangeable-lens cameras is competitive and demand is elastic. *An increase in the manufacturing costs of interchangeable-lens cameras will increase market revenues.*

Question 2 (6 points) True, False, Uncertain? [Explain your answer.] Mobile phones are sold in a perfectly competitive market and sell for \$1000 each. The elasticity of demand is -2.0 . The government recently passed a \$20 unit tax on mobile phones (roughly 2%), and the quantity demanded of mobile phones decreases by approximately 4.0%. *Supply for mobile phones is perfectly inelastic.*

Question 3 (6 points) True, False, Uncertain? [Explain your answer.] Suppose that when price is measured in \$/kilogram and demand is measured in kilograms, that the own-price elasticity of demand is $\varepsilon^d = -0.2$. If instead of using dollars, we measured prices in Gitcoins, where 1 Gitcoin is equal to \$10, *the own-price elasticity of demand remains unchanged and equal to -0.2.*

Question 4 (6 points) True, False, Uncertain? [Explain your answer.] Consider the following quote from *Ars Technica*, December 1, 2009, and evaluate the truth of the italicized portion given the stated facts:

“Following a Department of Justice investigation, most of the major LCD makers admitted to engaging in price fixing efforts, and agreed to substantial fines.” Shortly afterward, on November 25, Nokia sued the LCD manufacturers claiming that their price fixing resulted in higher LCD component prices and reduced profits. “Given the number of phones Nokia shipped annually during the period where collusion has been documented, *Nokia undoubtedly was harmed by any price increase this cartel managed to create.*”

Question 5 (6 points) True, False, Uncertain? [Explain your answer.] The elasticity of demand for wheat is $\varepsilon^d = -0.5$, and this elasticity is constant. The supply of wheat is infinitely elastic (i.e., perfectly elastic). The current equilibrium price is $p^* = 5$ per bushel. The government proposes a new subsidy to wheat farmers equal to $s = 0.50$ per bushel. *The subsidy will increase the equilibrium consumption of wheat by approximately 2.5 percent.*

Short problems (150 points total)

Question 6 (14 points) Evelyn consumes only two goods: tea, t , and rice, r . Suppose that the prices of tea and rice both increase by 10 percent while Evelyn’s income remains constant.

- How do these price changes affect Evelyn’s budget constraint? Explain.
- How does Evelyn’s choice of the tea and rice change (i.e., do they increase, decrease, or does the answer require some unstated assumptions)? Explain.
- How does Evelyn’s marginal rate of substitution, $MRS_{t,r}$, at her new optimal consumption choice compare to her original marginal rate of substitution at her optimal choice before the price change? [You may assume her indifference curves are smooth and she buys positive amounts of both goods before and after the price changes.]

Question 7 (12 points) Waymond runs a small company that launders and presses shirts. He has a production function for processed shirts given by $f(x_1, x_2) = x_1 + 2x_2$, where x_1 and x_2 are two different types of labour inputs and the labour prices are $w_1 = 2$ and $w_2 = 8$.

- (a). Assuming that both x_1 and x_2 are flexible inputs, what is the optimal mixture of inputs to launder and press shirts?
- (b). How much does it cost Waymond to process 200 shirts?

Question 8 [18 points] Jobu is a price taker producing in a perfectly competitive market. Jobu's opportunity costs of producing in the short run and long run, respectively, are given by

$$C_{sr}^0(q) = \frac{1}{2}q^2.$$

$$C_{lr}^0(q) = 200 + 10q + \frac{1}{2}q^2.$$

(Note: the derivative of the short-run function is q and the derivative of the long-run function is $10 + q$.)

- (a). What is Jobu's short-run supply as a function of price? Either give the equation or draw Jobu's short-run supply curve precisely.
- (b). The industry in which Jobu competes is a constant-cost industry in which every firm (incumbents as well as potential entrants) have the same long-run opportunity cost function as Jobu. If the market demand function is $Q^d = 100 - 2p$, then what is the long-run market equilibrium price and quantity?
- (c). In the long-run equilibrium of (b), if the government subsidizes the firms by giving every seller \$10 per unit sold, what will be each firm's new marginal revenue (including the consumer payments and the government's subsidy) earned in equilibrium?

Question 9 (14 points) Keyser manages a firm which produces ceramic coffee mugs. His short-run total (opportunity) cost function for producing a case of coffee mugs is

$$C_{sr}^O(q) = 0.20q^2;$$

the derivative of this function is $0.4q$. Keyser is a price taker and can sell as many cases of coffee mugs as he likes at a market price of $p = \$200$.

- (a). What is Keyser's short-run profit-maximizing output? [You may assume that it is profitable for Keyser to operate.]
- (b). In addition to selling plain coffee mugs at a price of \$200, Keyser has a special customer, Verbal Industries, who is willing to pay \$300 per case for up to 100 cases of customized coffee mugs with its logo printed on each mug. Verbal Industries does not want more than 100 cases. It will cost Keyser an additional \$35 per case to add the personalized logo. That is, if Keyser produces a total of $q = q_p + q_c$ cases of mugs, of which q_p cases are plain and q_c cases are customized, then Keyser's total short-run opportunity cost of production is $C_{sr}^O(q)$ plus $35q_c$.

How many cases of each type of mug should Keyser produce and sell? Explain.

Question 10 (10 points) Dalmore sells a differentiated type of Whisky in a market that faces competition from generic Scotch Whisky and a Japanese Whisky (Hibiki). Dalmore's demand function is given by

$$q_D^d = 75 - 2p_D + p_S + 2p_H,$$

where p_D is the price of Dalmore, p_S is the price of generic (blended) Scotch, and p_H is the price of Hibiki. Dalmore believes that the price of generic Scotch will remain at $p_S = 25$ and the price of Hibiki will remain at $p_H = 100$, regardless of the price that Dalmore chooses. The constant unit opportunity cost of producing a bottle of Dalmore is $MC = AC = 20$. What is the profit-maximizing price for Dalmore's Scotch?

Question 11 (24 points) Alpha Inc. is a monopolist. The demand curve for its product is represented by the equation:

$$p = 250 - 2q.$$

Alpha currently has a single plant to produce output which has capacity of 40 units. The marginal and average cost of producing is 20, providing that it produces below capacity; it cannot produce more than 40 units.

- What price should Alpha Inc. set in order to maximize profits? Explain.
- Suppose the facts above are unchanged, but Alpha Inc. also has access to a second plant which has unlimited capacity, but has higher per-unit costs of 50. Denote the original plant as plant 1 (i.e., $C_1(q_1) = 20q_1$ for $q_1 \leq 40$) and the new facility as plant 2: $C_2(q_2) = 50q_2$ (with unlimited capacity).

What price should Alpha Inc. set in order to maximize profits? How much (if any) should it produce at each plant?

- Suppose that Alpha Inc. owns plant 1, but a competitor, Beta Inc., owns plant 2. Alpha Inc. and Beta both decide how much output to bring to market and the market price is determined by the demand relationship above (i.e., Alpha and Beta are playing a duopoly, Cournot-output game). Assume that Alpha finds it optimal to produce at capacity and bring $q_a^* = 40$ units to market.

How much output should Beta bring to the market?

Question 12 (30 points) McKenzie industries produces a beverage called Slurm and is currently a monopolist because no one can figure out the secret ingredient in Slurm (and hence no one can make a competing product in this market). McKenzie can produce up to 25 bottles of Slurm at a constant unit opportunity cost of $MC = AC = 2$ per unit (up to capacity). The demand curve for Slurm is

$$p = 100 - Q.$$

- How many bottles of Slurm should McKenzie sell in each period and at what price? Explain. (Note that McKenzie cannot produce more than 25 units.)
- What is the elasticity of demand at the optimal price in (a)?
- Fry discovers the secret ingredient in Slurm and figures out a way to produce a perfect substitute to what McKenzie sells. Fry's costs have a constant unit cost of $MC = AC = 5$ per unit, but he has unlimited capacity at this price.

If Fry thinks McKenzie will produce at capacity, $\hat{q}_m = 25$, how much should Fry supply to the market? Explain. [Hint: This problem is equivalent to a monopolist facing a competitive fringe supplying 25 units independent of market price.]

- Continuing with the facts in (c), is the optimal output you computed for Fry in (c) a Nash equilibrium? That is, is it optimal for McKenzie to produce at capacity, $q_m^* = 25$, providing that Fry produces according to your answer in (c)? Explain.

Question 13 (28 points) Gru is a monopolist toy maker who sells a very popular toy called a “Minion.” His constant marginal cost of producing a Minion is only \$4 per unit and he has no fixed costs. He sells to two distinct geographical markets (Freedonia and Metrocity). Gru can set different prices in the two markets without any problem of arbitrage because there is a large river separating them and there is no bridge. (Gru is the only person who has a boat capable of crossing the river.) Demand in Freedonia City is given by

$$q_f = 8 - \frac{1}{2}p_f$$

and demand in Metrocity is given by

$$q_m = 10 - \frac{1}{2}p_m.$$

Note that these demand relationships can also be written as $p_f = 16 - 2q_f^d$, and $p_m = 20 - 2q_m^d$.

- (a). What are Gru’s profit-maximizing monopoly prices in each of these markets: p_f^* and p_m^* ?
- (b). Unfortunately for Gru, before implementing the prices from part (a), a bridge is built connecting the two markets. Gru now realizes he must treat the two markets as one large market and set a single monopoly price.

What is his optimal price?

- (c). Gru’s minions are currently yellow in color. Gru does some market research and learns that consumers from Metrocity prefer yellow to any other color. In fact, for any different color, a Metrocity consumer has a willingness to pay that is exactly \$3 lower. Gru also discovers that Freedonia consumers are color blind and don’t care about the color of the minion toys. Furthermore, the color of the toy has no effect on Gru’s costs of production; he can just as cheaply produce a mixture of some yellow and some red minion toys.

Knowing all these facts, how can Gru increase his profit? Be specific, including colors and prices. [Hint: consider the difference in prices found in part (a).]

Suggested answers to examination

1 False. An increase in manufacturing costs will cause an inward shift in supply. Because demand is elastic, revenues decrease on net as the price and quantity move up the demand curve (i.e., the percentage increase in price less than offsets the percentage decrease in output).

2 False. Because the demand elasticity is -2 and the result of the tax is a 4 percent reduction in demand, we know that the buyer's after-tax price increased by roughly 2 percent. This implies that the tax was entirely passed on to the consumers. This is only possible when supply is perfectly elastic (i.e., elasticity of infinity), rather than perfectly inelastic (i.e., elasticity of 0).

Alternatively, you could note that if supply is perfectly inelastic, then the entire tax is absorbed by the sellers and the buyers continue to buy the same as before; this is inconsistent with the fact the demand for mobile phones decreases by 4 percent.

3 True. Elasticities are invariant to how one measures price or quantity. The own-price elasticity of demand remains unchanged at $\varepsilon^d = -0.2$ regardless of how we measure the monetary units.

4 Uncertain. If LCD component prices increased equally for all mobile phone manufacturers from the price fixing agreement, the increase in costs may have been entirely passed on to consumers (and indirectly) telecommunication companies that offer subsidized phones to market phone service contracts. This will be the case if the supply curve for mobile phones is perfectly elastic over the range of price changes.

5 False. Because the wheat supply is perfectly elastic, the supply curve is horizontal. Thus, any subsidy will be passed along to the consumers. This implies that price will decrease from $p_0^* = 5$ to $p_1^* = 5 - 0.50 = 4.50$. This is a 1/10th reduction in price, or roughly a 10 percent decrease. Using the elasticity of demand equal to -0.5 , this would translate into an increase in consumption equal to $(-10)(-0.5) = 5$ percent.

6 (a). Graphically, the budget constraint shifts inward in a parallel fashion, so that the slope of the budget line is unchanged.

[figure to be completed]

(b). They both decrease in demand if they are normal goods. Effectively, income has been reduced (i.e., the new budget line is mathematically identical to a budget line in which income decreased by 10 percent). If one of them is inferior, then that one would increase in demand while the normal good would decrease in demand.

(c). The new marginal rate of substitution remains the same at the optimum, since the price ratio has not changed. (Note, it was fine to further note that this may not always be the case if a corner solution emerges or the indifference curves have kinks in them, but you had to get the basic point to get full marks.)

7 (a). These inputs are perfect substitutes. Given that $\frac{w_1}{w_2} = 0.25$ and $\frac{MP_1}{MP_2} = 0.5$, Waymond should produce using only x_1 .

(b). Given Waymond is using only x_1 to clean shirts, he will need 200 units of x_1 to process 200 shirts. The total cost will be the cost of producing in this manner is $C = w_1 \cdot x_1^* = 2 \cdot 200 = 400$.

8 (a). The marginal output rule requires that q^* satisfies $p = MC(q^*)$ if the firm produces any output. Thus, $MR = 10$ equals $MC_{sr} = q$. Hence, $q = p$ is the output which satisfies Jobu's short-run marginal-output rule. Because $AC_{sr}^0(q) = \frac{1}{2}q$, it follows that the minimum average cost arises at $q = 0$. We conclude that her short-run supply function is a line starting from the origin with slope 1 (i.e., the supply curve equation is $p = q^s$ and, alternatively, the supply function equation is $q^s = p$.)

(b). In a constant-cost industry, the long-run market supply curve is a horizontal line with vertical intercept equal to the minimum AC_{lr}^O of each firm. To find the latter, we solve for $MC_{lr}(\hat{q}) = AC_{lr}^O(\hat{q})$:

$$10 + \hat{q} = \frac{200}{\hat{q}} + 10 + \frac{1}{2}\hat{q}.$$

Simplifying,

$$\frac{1}{2}\hat{q} = \frac{200}{\hat{q}},$$

so $\hat{q}^2 = 400$ or $\hat{q} = 20$. Substituting this into either long-run marginal or average cost yields $\min AC_{lr}^O = 30$. Hence, the supply curve is horizontal at $p = 30$. We can therefore conclude that for any demand curve, the equilibrium price is $p^* = 30$ if some firms operate. Using $p = 30$ in the demand function, we obtain $Q^* = 100 - 2(30) = 40$.

(c). With the government subsidy, the long-run minimum average cost of each firm decreases from 30 to 20. Thus, the equilibrium price is $p^* = 20$ and the firm receives $p_s = p^* + 10 = 30$. Alternatively, you could have noted that because supply is horizontal (i.e., infinitely elastic), the entire subsidy will be passed along to the consumer, $\Delta p_s = 0$, and thus the equilibrium seller price in (b) will be the same as p_s in (c).

9 (a). Setting $MR = MC_{sr}$, we have $200 = 0.4q$. Thus, $q^* = 500$ is the optimal quantity if any is produced at all.

[Because I said you may assume that Keyser should operate, you did not need to check the shutdown rule. In case you were wondering, the average opportunity cost of a case of mugs is $AC_{sr}^O = 0.2(500)^2/500 = 100 < p = 200$, so the assumption that Keyser operates is correct.]

(b). Notice that Keyser earns $\$100 - \$35 = \$65$ more profit per unit selling customized mugs to the special customer than he does selling a generic coffee mug to the competitive plain-mug market. Thus, if Keyser produced only 100 cases, he would sell them all to the special customer. Any additional mugs in excess of 100 cases will be sold on the coffee mug market at a price of \$200 per case. Notice that the marginal revenue of every case in excess of 100 has not changed. Thus, setting $MR = MC$, Keyser continues to make exactly as many cases of coffee mugs as in (a), except that now 100 of the 500 cases are customized and 400 are plain.

10 First, we determine the residual demand curve for Dalmore whisky, given the expected prices of generic Scotch and Hibiki.

$$q_D^d = 75 - 2p_D + (25) + 2(100),$$

or

$$q_D^d = 300 - 2p_D.$$

Inverting to obtain the demand curve, we have

$$p = 150 - \frac{1}{2}q_D.$$

It follows that the marginal revenue function for Dalmore is

$$MR(q_D) = 150 - q_D.$$

Setting MR equal to MC , we have

$$150 - q_D = 20,$$

and so $q_D^* = 130$. Substituting this quantity into Dalmore's demand curve, we obtain the optimal price of

$$p^* = 150 - \frac{1}{2}(130) = 85.$$

11 (a). First, we determine marginal revenue. Given demand is $p = 250 - 2q$, $MR(q) = 250 - 4q$ (same intercept as demand, but twice the slope). For $q \leq 40$, $MR(q)$ decreases from 250 to $250 - 4(40) = 90$. Thus, $MR(q) > MC = 20$ for $q \leq 40$. Thus, Alpha Inc. will want to produce up to capacity. In this case, $q^* = 40$.

(b). With two plants available for production, Alpha Inc. will always want to produce using the cheapest method. For $q \leq 40$, Alpha will want to produce using only plant 1. Its marginal cost in this case is 20. For any output above 40, Alpha must use the more expensive plant 2. Its marginal cost now increases to 50. We conclude that $MC = 20$ for $q < 40$ and $MC = 50$ for $q > 40$. Now consider marginal revenue. At $q = 40$, $MR(40) = 250 - 4(40) = 90$. Thus, $MR(q) > MC$ for q slightly above 40. It thus follows that Alpha will want to produce some units at the high-cost plant. Given MC is now 50, the optimal total output to produce is where $MR(q) = 50$. Thus,

$$250 - 4q = 50,$$

or $q^* = 50$. Of these 50 units, $q_1^* = 40$ and $q_2^* = 10$. The optimal price is $p^* = 250 - 2(50) = 150$.

(c). There are two possibilities in equilibrium: either Alpha produces at capacity, $q_a = 40$, or it produces at a lower level. We are told that Alpha finds it optimal to produce at capacity. Thus, we need only determine what Beta will want to produce. Beta's residual demand curve is

$$p = 250 - 2(40 + q_b) = 170 - 2q_b.$$

Thus, marginal revenue for Beta is

$$MR_b(q_b) = 170 - 4q_b.$$

Setting this equal to $MC_b = 50$, we obtain

$$170 - 4q_b = 50,$$

or $q_b^* = 30$. Given Beta brings 30 units to market, the market price will be

$$p^* = 250 - 2(40 + 30) = 110.$$

12 (a). First consider McKenzie's marginal revenue function:

$$MR(q_m) = 100 - 2q_m.$$

At a marginal cost of 2, McKenzie would like to produce

$$100 - 2q_m = 2,$$

or $q_m = 49$ units. He can only supply at capacity of 25, so the optimal amount to supply is $q_m^* = 25$. At this output, $MR_m = 50$ which greatly exceeds marginal cost, but he can't produce any more. Selling just $q_m^* = 25$ units allows McKenzie to charge $p^* = 100 - 25 = 75$ per unit.

(b). The formula to use is

$$\varepsilon^d = \frac{dQ^d}{dp} \frac{p^*}{Q^*} = (-1) \frac{75}{25} = -3.$$

(c). Suppose that $q_m = 25$. Fry's residual demand curve is given by

$$p = 100 - (q_f + 25) = 75 - q_f.$$

With this demand, Fry's marginal revenue is

$$MR_f(q_f) = 75 - 2q_f.$$

Setting this equal to $MC_f = 5$,

$$75 - 2q_f = 5,$$

so $q_f^* = 35$. Given $Q^* = 25 + 35 = 60$, the market price is $p^* = 100 - 60 = 40$.

(d). Given that Fry is choosing $q_f^* = 35$, we can compute McKenzie's residual demand curve:

$$p = 100 - (35 + q_m) = 65 - q_m.$$

Thus, McKenzie's marginal revenue is

$$MR_m(q) = 65 - 2q_m.$$

Note that at $q_m = 25$, this marginal revenue is 15, which exceeds marginal cost. Hence, McKenzie can do no better than produce at capacity. $q_m^* = 25$ and $q_f^* = 35$ is a Nash equilibrium to the Cournot duopoly game.

13 (a). Because there are no fixed costs, Gru's minimum average cost is zero and so he will always operate if he can sell above marginal cost (which she can). Thus, we apply the marginal-output rule to each market. Be careful here if you use the trick that MR has the same intercept as the demand curve with twice the slope – you need to use the relationships with price on the left-hand side.

$$MR_f(q_f) = 16 - 4q_f = MC = 4,$$

$$MR_m(q_m) = 20 - 4q_m = MC = 4.$$

Solving for the optimal outputs, we obtain $q_f^* = 3$ and $q_m^* = 4$. The corresponding optimal prices are $p_f^* = 16 - 2q_f^* = 10$ and $p_m^* = 20 - 2q_m^* = 12$.

(b). With only one market, we need to find the aggregate demand. In this case, we simply add up the demands of each market:

$$Q^d = q_f^d + q_m^d = \left(8 - \frac{1}{2}p\right) + \left(10 - \frac{1}{2}p\right) = 18 - p.$$

Inverting to obtain the aggregate demand curve equation, we obtain

$$p = 18 - Q.$$

Setting $MR = MC$, we have

$$MR(Q) = 18 - 2Q = MC = 4,$$

and so

$$Q^* = 7.$$

The corresponding optimal price is $p^* = 18 - 7 = 11$.

(c). Gru can effectively implement the solution in part (a) by selling his toys in two different colors. In particular, he can sell yellow minion toys at $p_y^* = 12$ and his red toys at $p_r^* = 10$ and the consumers from each city will buy just as they did in part (a). Specifically, because Metrocity consumers dislike red by \$3 per toy, consumers who would have purchased the toy in part (a) will buy the yellow toy at 12 rather than the cheaper red toy at 10. Freedonia buyers are color blind, so they will buy the cheapest toy, the red one priced at 10. Thus, the same number of Freedonia consumers purchase as under (a).